ACM/IDS 104 APPLIED LINEAR ALGEBRA
PROBLEM SET 6: SOLUTIONS

KONSTANTIN M. ZUEV

Problem 1. (10 points) The Power Method (aka von Mises Iteration)
Complete Problem 1 in `PS6.mlx`.

Solution: See `PS6solutions.mlx` (uploaded to the Piazza page).

Problem 2. (10 points) Ranking US Airports using PageRank
Complete Problem 2 in `PS6.mlx`.

Solution: See `PS6solutions.mlx` (uploaded to the Piazza page). The ID numbers of 10 most important airports are

Rank	1	2	3	4	5	6	7	8	9	10
$\alpha = 0.10$	6	1	7	3	2	21	11	8	18	10
$\alpha = 0.15$	6	7	3	1	2	21	11	8	10	18
$\alpha = 0.20$	6	7	3	1	2	21	11	10	8	18

Problem 3. (10 points) Symmetric Matrices
In this problem, we will practice and explore different aspects of symmetric matrices, which form one of the most important classes of matrices in applications. To start, let's practice diagonalization.

(a) (2 points) Diagonalize the following matrix (without using numerical software packages)

$$A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 5 & 0 \\ 4 & 0 & 1 \end{bmatrix}. \tag{1}$$

That is, find a diagonal matrix Λ and an orthogonal matrix Q such that $A = Q\Lambda Q^T$.

Now, let $\mathrm{Sym}(n)$ denote the set of all real symmetric matrices of size $n \times n$.

(b) (2 points) Is $\mathrm{Sym}(n)$ a vector space with respect to the standard matrix operations?
(c) (2 points) What is the dimension of $\mathrm{Sym}(n)$? Construct a basis of $\mathrm{Sym}(n)$.
(d) (2 points) Find the orthogonal complement $\mathrm{Sym}(n)^\perp$ to $\mathrm{Sym}(n) \subset \mathbb{M}_{n\times n}$ with respect to the Frobenius inner product $\langle A, B\rangle_F = \mathrm{tr}(A^T B)$, introduced in Practice Problem 4B.
(e) (2 points) For symmetric matrices, the Frobenius inner product reduces to

$$\langle A, B\rangle_F = \mathrm{tr}(AB), \qquad A, B \in \mathrm{Sym}(n). \tag{2}$$

Prove that the Frobenius norm induced by this inner product satisfies

$$\|A\|_F^2 = \sum_{i=1}^{n} \lambda_i^2, \tag{3}$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $A \in \mathrm{Sym}(n)$.

Solution:

(a) Since A is symmetric, the spectral theorem guarantees that the decomposition $A = Q\Lambda Q^T$ exists. The matrix $\Lambda = \mathrm{diag}\{\lambda_1, \lambda_2, \lambda_3\}$ is a diagonal matrix of eigenvalues of A, and the matrix $Q = [q_1, q_2, q_3]$ is an orthonormal matrix of its eigenvectors, $Aq_i = \lambda_i q_i$.

E-mail address: `kostia@caltech.edu`.

The eigenvalues are the roots of the characteristic polynomial:

$$\begin{aligned} p_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 0 & 4 \\ 0 & 5-\lambda & 0 \\ 4 & 0 & 1-\lambda \end{bmatrix} &= (1-\lambda)^2(5-\lambda) - 4^2(5-\lambda) \\ = (5-\lambda)((1-\lambda)^2 - 4^2) &= -(5-\lambda)^2(3+\lambda). \end{aligned} \tag{4}$$

So, the eigenvalues are

$$\lambda_1 = -3, \quad \text{and} \quad \lambda_2 = \lambda_3 = 5. \tag{5}$$

To find an eigenvector v_i corresponding to the eigenvalue λ_i, we need to solve $(A - \lambda_i I)v_i = 0$. Note that $\lambda_2 = \lambda_3 = 5$ is a repeated eigenvalue. To construct Q, we need to find not just two arbitrary linear independent eigenvectors v_2 and v_3 corresponding to $\lambda_2 = \lambda_3 = 5$, but two orthogonal eigenvectors[1].

$$\begin{aligned} \lambda_1 = -3: \quad & (A - \lambda_1 I)v_1 = 0 \quad \Leftrightarrow \quad \begin{bmatrix} 4 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 4 \end{bmatrix} v_1 = 0 \quad \Leftrightarrow \quad v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}. \\ \lambda_{2,3} = 5: \quad & (A - \lambda_{2,3} I)v_{2,3} = 0 \quad \Leftrightarrow \quad \begin{bmatrix} -4 & 0 & 4 \\ 0 & 0 & 0 \\ 4 & 0 & -4 \end{bmatrix} v_{2,3} = 0 \quad \Leftarrow \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}. \end{aligned} \tag{6}$$

The columns q_1, q_2, q_3 of matrix Q are the normalized eigenvectors $q_i = v_i/\|v_i\|$. So, the spectral decomposition of A looks as follows:

$$\underbrace{\begin{bmatrix} 1 & 0 & 4 \\ 0 & 5 & 0 \\ 4 & 0 & 1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}}_{Q^T} \tag{7}$$

(b) Yes, $\mathrm{Sym}(n)$ is a vector space, since $\mathrm{Sym}(n)$ is a subset of the set of all $n \times n$ matrices, $\mathrm{Sym}(n) \subset \mathbb{M}_{n\times n}$, and $\mathrm{Sym}(n)$ is closed under the standard matrix operations:

$$A, B \in \mathrm{Sym}(n),\ \alpha \in \mathbb{R} \implies A + B \in \mathrm{Sym}(n) \quad \text{and} \quad \alpha A \in \mathrm{Sym}(n). \tag{8}$$

Indeed:

$$(A+B)^T = A^T + B^T = A + B \quad \text{and} \quad (\alpha A)^T = \alpha A^T = \alpha A. \tag{9}$$

Therefore, $\mathrm{Sym}(n)$ is a subspace of $\mathbb{M}_{n\times n}$, and hence a vector space itself.

(c) The dimension of $\mathrm{Sym}(n)$ is $n(n+1)/2$, since the following collection of matrices forms a basis of $\mathrm{Sym}(n)$:

- D_i is a diagonal matrix with the (i,i)-element equal to 1 and the remaining elements equal to zero, where $i = 1, \ldots, n$.
- S_{ij} is a symmetric matrix with the (i,j)- and (j,i)-elements equal to 1 and the remaining elements equal to zero, where $i < j$.

These $n + n(n-1)/2 = n(n+1)/2$ matrices form a basis because any symmetric matrix can be written uniquely as a linear combination of these matrices.

For example, for $n = 2$

$$D_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad S_{12} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{10}$$

[1]Alternatively, we can find any two linear independent eigenvectors v_2 and v_3 and then orthogonalize them using the Gram-Schmidt process.

For $n = 3$,

$$\begin{aligned} D_1 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad D_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ S_{12} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad S_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad S_{23} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}. \end{aligned} \tag{11}$$

(d) The orthogonal complement $\mathrm{Sym}(n)^\perp$ to $\mathrm{Sym}(n) \subset \mathbb{M}_{n\times n}$ with respect to the Frobenius inner product consists of all $n \times n$ matrices $B \in \mathbb{M}_{n\times n}$ such that $\langle A, B\rangle_F = 0$ for all symmetric matrices $A \in \mathrm{Sym}(n)$. Equivalently, $\mathrm{Sym}(n)^\perp$ consists of all $n \times n$ matrices $B \in \mathbb{M}_{n\times n}$ orthogonal to the basis vectors $\{D_i, S_{ij}\}$ of $\mathrm{Sym}(n)$ constructed in part (c). For symmetric A,

$$\langle A, B\rangle_F = \mathrm{tr}(A^T B) = \mathrm{tr}(AB) = \sum_{i=1}^{n} (AB)_{ii} = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} A_{ij} B_{ji} \right) = \sum_{i,j=1}^{n} A_{ji} B_{ji}. \tag{12}$$

Therefore, $B \in \mathrm{Sym}(n)^\perp$ if and only if

$$\begin{aligned} \langle D_i, B\rangle_F = 0 \quad &\Leftrightarrow \quad B_{ii} = 0, \quad i = 1, \ldots, n, \\ \langle S_{ij}, B\rangle_F = 0 \quad &\Leftrightarrow \quad B_{ij} + B_{ji} = 0, \quad i < j. \end{aligned} \tag{13}$$

This means that B is a skew-symmetric matrix. So,

$$\mathrm{Sym}(n)^\perp = \mathrm{Skew}(n). \tag{14}$$

This result provides an alternative proof of Problem 4(c) in PS1, where you were asked to show that every square matrix A can be uniquely written as the sum of a symmetric and a skew-symmetric matrix.

(e) Since A is symmetric, we can use its spectral decomposition:

$$A = Q\Lambda Q^T, \tag{15}$$

where Q is orthogonal and $\Lambda = \mathrm{diag}\{\lambda_1, \ldots, \lambda_n\}$. Then:

$$\|A\|_F^2 = \mathrm{tr}(A^2) = \mathrm{tr}(Q\Lambda Q^T Q\Lambda Q^T) = \mathrm{tr}(Q\Lambda^2 Q^T) = \mathrm{tr}(\Lambda^2 Q^T Q) = \mathrm{tr}(\Lambda^2) = \sum_{i=1}^{n} \lambda_i^2. \tag{16}$$

Here we used the identity $\mathrm{tr}(AB) = \mathrm{tr}(BA)$, which we proved in Practice Problem 14B(a).

Problem 4. (10 points) Principal Component Analysis
Complete Problem 4 in `PS6.mlx`.

Solution: See `PS6solutions.mlx` (uploaded to the Piazza page). The covariance matrices of X and Y, and principal components are shown below.

$$C_X = \begin{bmatrix} 0.98 & -0.68 \\ -0.68 & 1.951 \end{bmatrix} \quad C_Y = \begin{bmatrix} 0.63 & 0 \\ 0 & 2.3 \end{bmatrix} \quad v_1 = \begin{bmatrix} -0.46 \\ 0.89 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0.89 \\ 0.46 \end{bmatrix} \tag{17}$$

Problem 5. (10 points) Spectral Method for Graph Partitioning
Complete Problem 5 in `PS6.mlx`.

Solution: See `PS6solutions.mlx` (uploaded to the Piazza page). The two candidate partitions, s_+ and s_-, are shown in Figure 2. The corresponding cut sizes are $R_+ = 3$ and $R_- = 200$. Thus s_+ (left panel) is the target solution, which is of course the best possible partition available for graph G.

Acknowledgements. A huge thanks to Pavlos Stavrinides for editing my clumsy MATLAB m-files and converting them into neat live scripts!

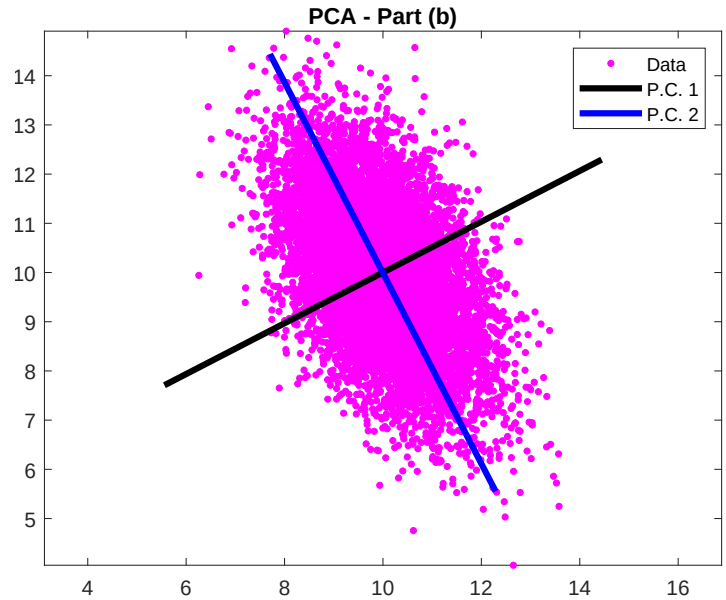


FIGURE 1. Principal Components.

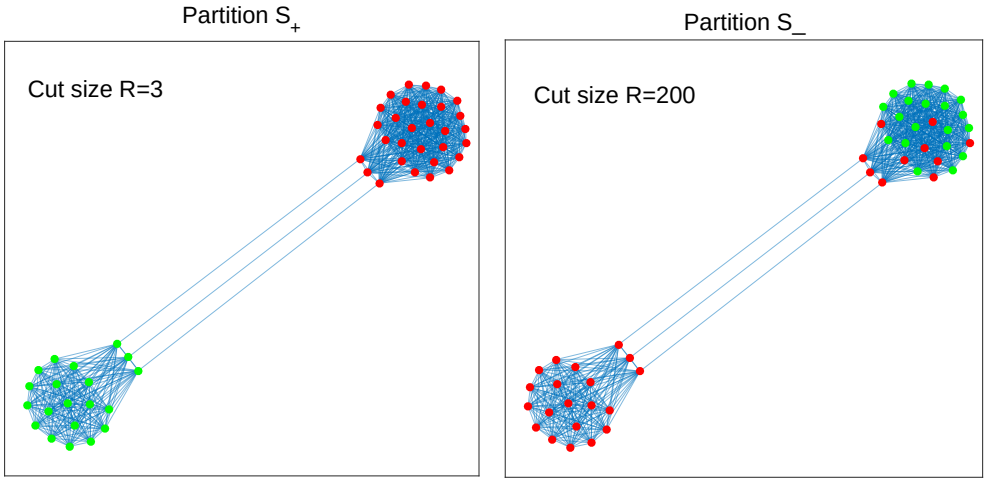


FIGURE 2. Two candidate partitions.

ACM/IDS 104 - Problem Set 6 - MATLAB Solutions

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

NOTE: Start with Problem 1, at the *bottom of this livescript*. (It's at the bottom because it's a MATLAB live function)

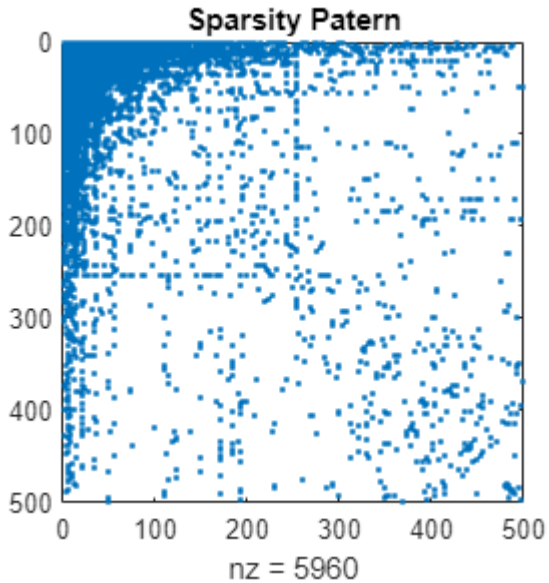
Problem 2 (10 points) Ranking US Airports using PageRank

The PageRank algorithm can be used for ranking not only web pages, but any "entities" organized into a network. The `USAirTransportation.mat` file contains the adjacency matrix A for the network of (anonymized) US airports: nodes represent airports and links represent air travel connections among them. That is, airports i and j are connected $(A_{ij} = 1)$ if there are direct flights between them. The network in `USAirTransportation.mat` is obtained by considering the 500 US airports with the largest amount of traffic from publicly available data. Implement the PageRank algorithm (see lecture 12) for ranking the airports. Use your function from Problem 1 for finding the importance score vector. What are the ID numbers of 10 most important airports for $\alpha = 0.10, 0.15, 0.20$?

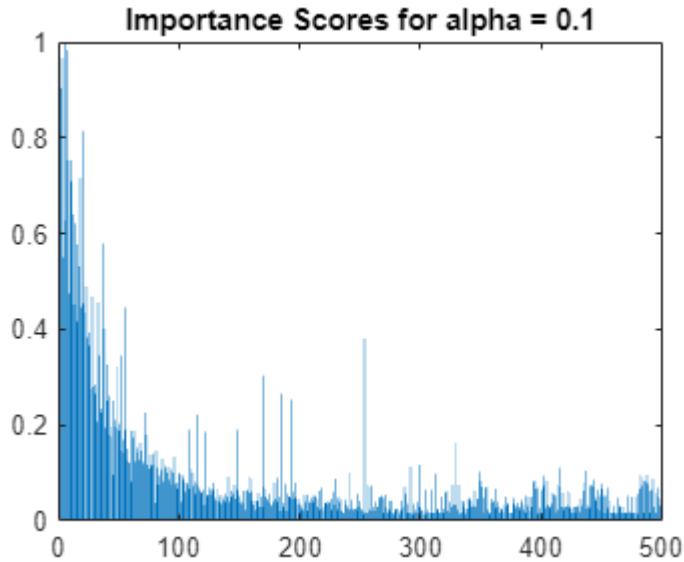
Functions that may be useful for this problem:

`repmat(), sort(), disp(), strcat(), num2str()`

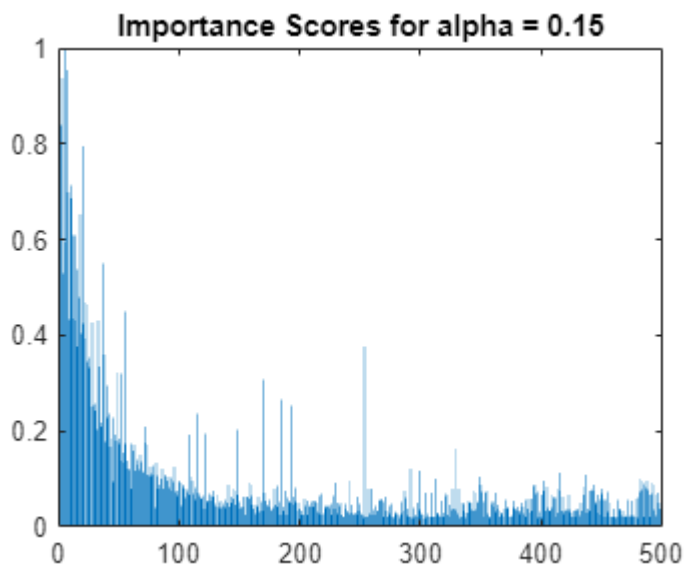
```
load USAirTransportation.mat % dataset
%{
Let us see how matrix A looks like
%}
figure;
spy(A);
title('Sparsity Patern');
```



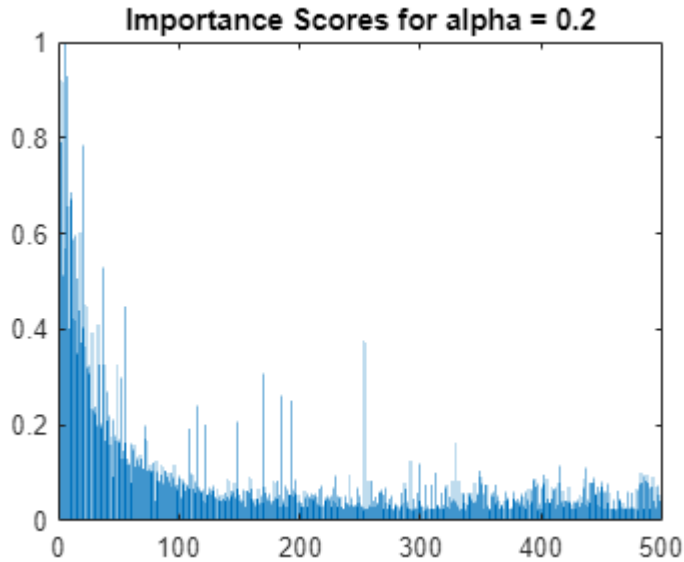

```
alpha = [0.10 0.15 0.20]; % values of alpha
[~,n]=size(A); % size of A
%{
Implement the algorithm below
You can use bar() to visualize the scores
Report the top 10 airports (in descending order) for each alpha
%}
L=A./repmat(sum(A),n,1);  % link matrix
for i = alpha
LL=(1-i)*L+i*repmat(ones(1,n)/n,n,1); % LL is \tilde{L}
[lambda,x]=power_method(LL);
% lambda must be 1, v is the eigenvector of importance scores
figure;
bar(x);
title(strcat("Importance Scores for alpha = ", num2str(i)));
[x,ind]=sort(x,'descend');
disp(strcat("The Top 10 airports for alpha = ", num2str(i)));
disp(ind(1:10));
end
```




```
The Top 10 airports for alpha = 0.1
     6
     1
     7
     3
     2
    21
    11
     8
    18
    10
```




```
The Top 10 airports for alpha = 0.15
     6
     7
     3
     1
     2
    21
    11
     8
    10
    18
```

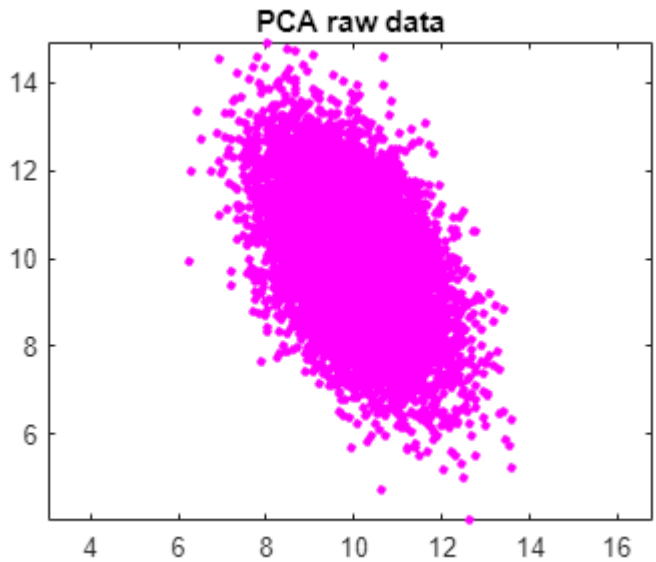



```
The Top 10 airports for alpha = 0.2
     6
     7
     3
     1
     2
    21
    11
    10
     8
    18
```

Problem 4 (10 points) Principal Component Analysis

In this problem, you will find the principal components of the dataset `pca_data.mat`, which contains the $m \times n$ matrix X. Columns of X correspond to $n = 10^4$ measurements and rows of X correspond to $m = 2$ quantities (features). Let us see what the data looks like:

```
load pca_data.mat
figure;
plot(X(1,:),X(2,:),'.m','MarkerSize',10);
title("PCA raw data");
axis equal
```



Part (a)

Find the $m \times m$ covariance matrix of X. Use `disp()` to display it.

```
[m, n] = size(X); % m = # measurement types (features); n = # of measurements
for i = 1 : m
    mu(i) = mean(X(i, :));
    Xn(i, :) = X(i, :) - mu(i); % normalization
end
Cx = Xn*Xn'/n; % Covariance matrix
%{
Or, equivelantly:
Cx=cov(X');  % formula in line 32 is valid only if row means are zero.
%}
disp("The covariance matrix of X is:");
```

```
The covariance matrix of X is:
```

```
disp(Cx);
```

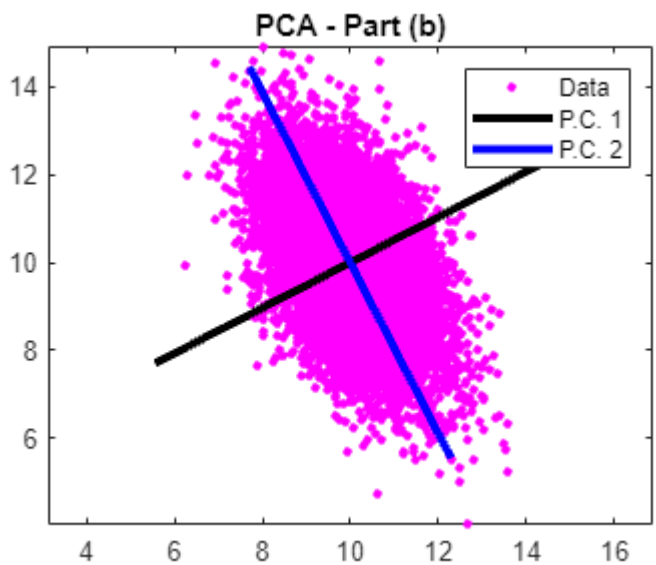
```
    0.9829   -0.6808
   -0.6808    1.9508
```

Part (b)

Find the m principal components of X and show them on top of the original data. That is, if v is a principal component, then plot the line $l(t) = \mu + vt$ over the scatter plot shown above, where $\mu \in \mathbb{R}^2$ is the centroid of the data.

```
%{
Find the eigenvectors
Normalize
Obtain the centroids
Create the lines
Plot
```

```
%}
[V1, D] = eig(Cx); % finding eigenstuff
v_1 = V1(:, 1); % eigenvector 1
v_1 = v_1 / norm(v_1); % normalize
v_2 = V1(:, 2); % eigenvector 2
v_2 = v_2 / norm(v_2); % normalize
c_1 = mu(1); % centroid 1
c_2 = mu(2); % centroid 2
p_1 = @(t) c_1 + v_1(1) * t; % get x^1 coord. for point p
q_1 = @(t) c_2 + v_2(1) * t; % get x^1 coord. for point q
p_2 = @(t) c_1 + v_1(2) * t; % get x^2 coord. for point p
q_2 = @(t) c_2 + v_2(2) * t; % get x^2 coord. for point q
% PLOT
plot(X(1,:),X(2,:),'.m','MarkerSize',10);
hold on
fplot(p_1, p_2, "-k", "lineWidth", 3);
fplot(q_1, q_2, "-b", "lineWidth", 3);
title("PCA - Part (b)");
legend("Data","P.C. 1", "P.C. 2");
axis equal
```



Part (c)

If we change the basis of $\mathbb{R}^m$ from the standard basis to the basis of the principal components $[v_1, \cdots, v_m] = V$ of X, the data matrix will transform to $Y = V^{-1}X$. Find the covariance of the transformed data Y. Use `disp()` to display it.

```
Y = V1'*X;
for i = 1 : m
    Yn(i,:) = Y(i,:)-mean(Y(i,:)); % normalizatoin
end
Cy = Yn*Yn'/n; % = Lambda
%{
```

```
Or, equivelantly:
Cy=cov(Y'); % formula in line 61 is valid only of row means are zero.
%}
disp("The covariance matrix of the transformed data Y is:");
```

```
The covariance matrix of the transformed data Y is:
```

```
disp(Cy);
```

```
    0.6316    0.0000
    0.0000    2.3021
```

Part (d)

Principal component analysis is a very popular data analysis technique and it is often used in various applications. As such, it is implemented in most numerical software packages. In MATLAB, this is a built-in function `pca()`. In the simplest form, `V = pca(X)` returns the principal components of the data matrix X, where rows of X correspond to the *measurements* and columns correspond to the *features*. Find the principal components of X using `pca()` and compare them with the ones found in part (b). Are they the same?

```
V2 = pca(X');
disp("Found in part (b):");
```

```
Found in part (b):
```

```
disp(V1);
```

```
   -0.8886   -0.4586
   -0.4586    0.8886
```

```
disp("Using pca():");
```

```
Using pca():
```

```
disp(V2);
```

```
   -0.4586    0.8886
    0.8886    0.4586
```

```
disp("The vectors agree!");
```

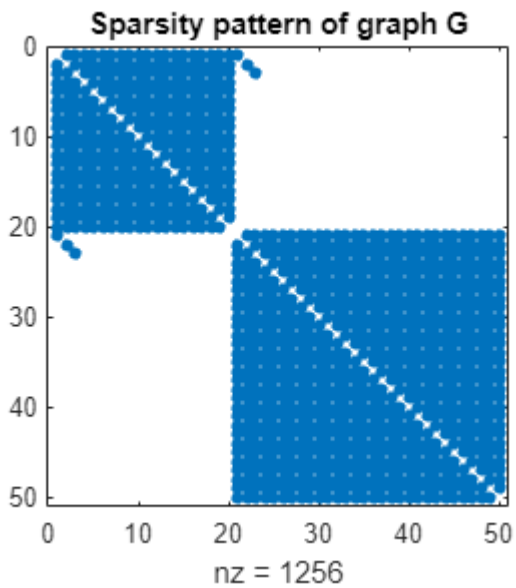
```
The vectors agree!
```

Problem 5 (10 points) Spectral Method for Graph Partitioning

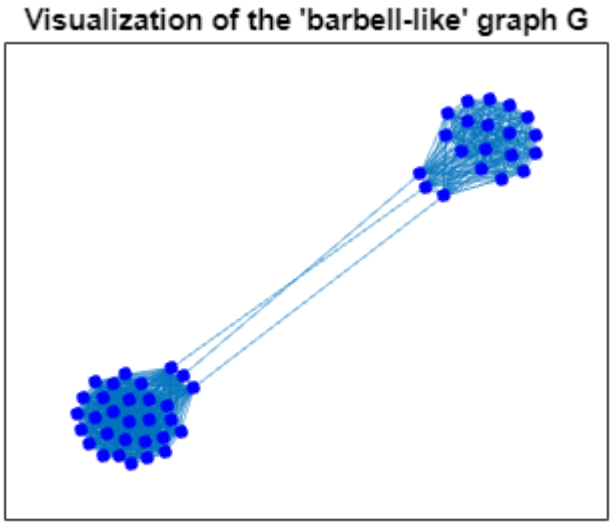
A complete graph K_n with n vertices is a graph in which each pair of graph vertices is connected by an edge. Let G be a barbell-like graph obtained by connecting K_{n_1} and K_{n_2} with $m = 3$ bridges, where $n_1 = 20$ and $n_2 = 30$. The graph G and the sparsity pattern of its adjacency matrix are shown:

```
%{
Adjacency matrix
%}
n1=20;
n2=30;
```

```
B1=ones(n1,n1)-diag(ones(1,n1));
B2=ones(n2,n2)-diag(ones(1,n2));
B=[B1, zeros(n1,n2);zeros(n2,n1), B2]; % adjacency matrix B of graph G
for i=1:3
    B(i,n1+i)=1; B(n1+i,i)=1; % bridges
end
%{
Visualization
%}
figure;
spy(B);
title("Sparsity pattern of graph G");
```




```
figure;
G = graph(B);
plot(G,'NodeLabel',{},'NodeColor','b');
title("Visualization of the 'barbell-like' graph G");
```



Use the spectral partitioning method to find a division of the graph G into two subgraphs of size $n_1 = 20$ and $n_2 = 30$ such that the number of edges between the subgraphs (the cut size) is minimized. Visualize the obtained partition and find the corresponding cut size. Useful functions for this problem:

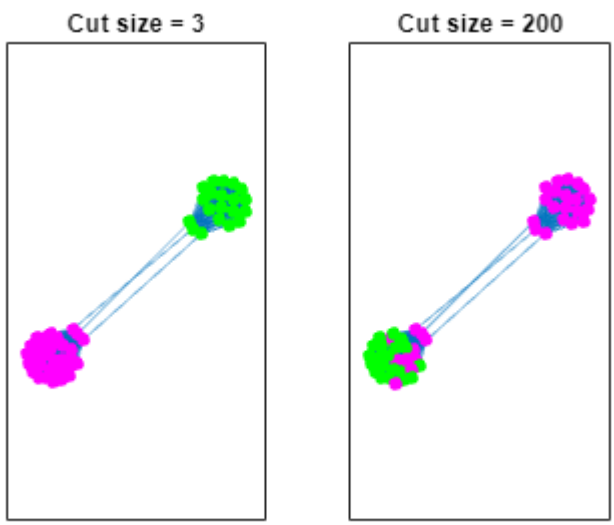
`eigs(), find(), highlight(), plot()`

```
%{
-> Find the graph Laplacian
-> Find the Fiedler vector
-> Determine the 2 candidate partitions
-> Visualize
-> Report the cut size using disp()
%}

% LAPLACIAN & FIEDLER VECTOR
La = diag(sum(B))-B;  % or simply La=laplacian(G)
[V,Lambda] = eigs(La,2,'sm'); % two smallest eigenvalues and
```

```
Warning: First input matrix is close to singular or badly scaled (RCOND =  9.177844e-18) and results may be
inaccurate. Consider specifying a small nonzero numeric sigma value instead of 'smallestabs' to improve the
condition of the matrix.
```

```
                                % the corresponding eigenvectors
v2=V(:,2); % the Fiedler vector (corresponds to the 2nd smallest eigenvalue)

% PARTITIONS
[~,ind]=sort(v2);
Sp=-ones(n1+n2,1);
Sp(ind(n2+1:end))=1; % s "plus"
Rp=Sp'*La*Sp/4; % the corresponding cut size
Sm=-ones(n1+n2,1);
Sm(ind(1:n1))=1; % s "minus"
Rm=Sm'*La*Sm/4; % the corresponding cut size
```

```
% VISUALIZATION
figure;
subplot(1,2,1);  % partion Sp
h=plot(G,'NodeLabel',{});  % Graph G
highlight(h,find(Sp>0),'NodeColor','g'); % subgraph 1
highlight(h,find(Sp<0),'NodeColor','m'); % subgraph 2
title(strcat("Cut size = ", num2str(Rp)));
axis equal
subplot(1,2,2);   % partition Sm
h=plot(G,'NodeLabel',{});  % Graph G
highlight(h,find(Sm>0),'NodeColor','g'); % subgraph 1
highlight(h,find(Sm<0),'NodeColor','m'); % subgraph 2
title(strcat("Cut size = ", num2str(Rm)));
axis equal
```




```
% REPORTING CUT SIZE
disp(strcat("The minimum cut size is ", num2str(Rp)));
```

```
The minimum cut size is 3
```

Problem 1 (10 points) - The Power Method (aka Von Mises Iteration)

Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \cdots, \lambda_n$ such that:

$$|\lambda_1| > |\lambda_2| > \cdots > |\lambda_n|$$

The eigenvalue λ_1 is called the dominant eigenvalue and the corresponding eigenvector v_1 is called the dominant eigenvector. Write a function that takes A as an argument and uses the Power method (Lecture 12) to compute λ_1 and v_1. We can check that λ_1 is correct by comparing it with the `eig()` function for a random matrix. An example is provided below:

```
%{
```

```
E = [9,8,7;6,5,4;3,2,1];
[b, d] = eig(E);
[J, VE] = power_method(E);
disp(d(1));
disp(J);
%}
```

```
function [lambda, v] = power_method(A)
[n, ~] = size(A); % size of nxn matrix A
v_prev = rand(n, 1); % random starting vector
tol = Inf; % relative error
%{
Implement the algorithm in the while loop
%}
while tol > 1e-15
    v = A * v_prev;
    lambda = norm(v, Inf);
    v = v / lambda;
    tol = norm(v - v_prev, Inf)/norm(v_prev, Inf);
    v_prev = v;
end
end
```